

1 Solution — GIAM 1.7

Problem 1

1.1 Problem

Consider the numbers from 1 to 10. Give the set of pairs of these numbers that corresponds to the divisibility relation.

1.2 Answer

Writing the relation as the set of ordered pairs (a, b) for which $a \mid b$ (read " a divides b "), with $a, b \in \{1, 2, \dots, 10\}$:

$$\begin{aligned} | = \{ & (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (1, 7), (1, 8), (1, 9), (1, 10), \\ & (2, 2), (2, 4), (2, 6), (2, 8), (2, 10), \\ & (3, 3), (3, 6), (3, 9), \\ & (4, 4), (4, 8), (5, 5), (5, 10), (6, 6), (7, 7), (8, 8), (9, 9), (10, 10) \}. \end{aligned}$$

There are $\boxed{27}$ pairs.

1.3 How the Pairs Are Found

Following the textbook's own convention — recall the less-than relation on $\{1, \dots, 5\}$ was listed as the pairs whose *first* entry is smaller than the second — we include (a, b) exactly when the first number divides the second. Going divisor by divisor:

a	the b with $a \mid b$	count
1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	10
2	2, 4, 6, 8, 10	5
3	3, 6, 9	3
4	4, 8	2

5	5, 10	2
6	6	1
7	7	1
8	8	1
9	9	1
10	10	1

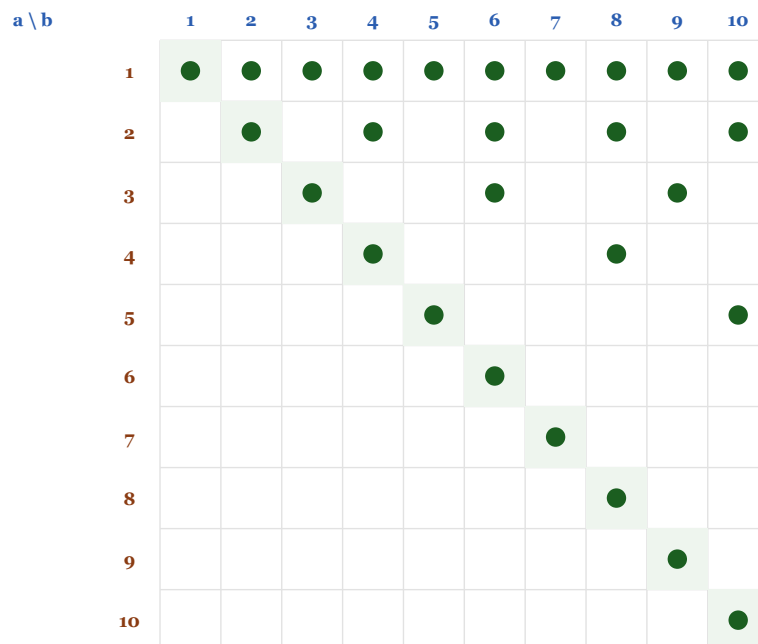
Table 1.1

Summing the counts: $10 + 5 + 3 + 2 + 2 + 1 + 1 + 1 + 1 + 1 = 27$.

1.4 The Relation as a Grid

Displaying the relation as a 10×10 grid makes its shape vivid — a dot sits at (a, b) precisely when $a \mid b$:

*Mark cell (a, b) when a divides b . A dot means the pair (a, b) is in the relation.
All 27 dots lie on or above the diagonal, since $a \mid b$ forces $a \leq b$.*



Row $a=1$ is full (1 divides everything); primes $6 < p \leq 10$ give only (p, p) ; the shaded diagonal is reflexivity $a \mid a$.

27 pairs in total = $10 + 5 + 3 + 2 + 2 + 1 + 1 + 1 + 1 + 1$.

Figure 1.1: A 10 by 10 grid, rows $a = 1..10$ and columns $b = 1..10$, with a green dot in cell (a,b) exactly when a divides b . Row 1 is completely full because 1 divides every number. Row 2 hits the even columns 2,4,6,8,10; row 3 hits 3,6,9; and so on. The primes 7 and larger, and 6,8,9,10 as first entry, contribute only the single diagonal pair (a,a) . The shaded main diagonal marks reflexivity a divides a . All 27 dots lie on or above the diagonal because a divides b forces a to be at most b . Total $27 = 10+5+3+2+2+1+1+1+1+1$.

1.5 Remark — Divisibility Is Reflexive, So the Diagonal Is Included

Unlike the strict less-than relation (whose set of pairs on $\{1,\dots,5\}$ *omits* $(1,1), (2,2), \dots$), divisibility is **reflexive**: every number divides itself, since $a = 1 \cdot a$. That is why all ten diagonal pairs $(1,1), (2,2), \dots, (10,10)$ appear in the answer. Watching whether a relation contains its diagonal is the quickest visual test for reflexivity — and it is the single biggest way this list differs from the “ $<$ ” example the section gives.

1.6 Remark — Why Every Dot Sits On or Above the Diagonal

No pair (a,b) with $a > b$ can appear, because a positive divisor of b can be at most b itself: if $a \mid b$ and $b > 0$ then $a \leq b$. So the relation lives entirely in the “upper triangle” of the grid. This makes divisibility an **antisymmetric** relation on the positive integers — if $a \mid b$ and $b \mid a$ both hold, then $a \leq b$ and $b \leq a$, forcing $a = b$. Reflexive + anti-symmetric + transitive ($a \mid b$ and $b \mid c \Rightarrow a \mid c$) makes divisibility a **partial order**, which is exactly why it can be drawn as the divisibility “lattice” the book returns to in Chapter 6.

1.7 Remark — The Row Lengths Are the Counting Function $\lfloor 10/a \rfloor$

The number of multiples of a in $\{1,\dots,10\}$ — the length of row a — is exactly $\lfloor 10/a \rfloor$:

$$\left\lfloor \frac{10}{1} \right\rfloor, \left\lfloor \frac{10}{2} \right\rfloor, \dots, \left\lfloor \frac{10}{10} \right\rfloor = 10, 5, 3, 2, 2, 1, 1, 1, 1, 1,$$

tying this problem back to the floor/quotient material of Section 1.4. Summing $\sum_{a=1}^{10} \lfloor 10/a \rfloor = 27$ counts *all* (divisor, multiple) pairs at once — equivalently it counts $\sum_{b=1}^{10} d(b)$, the total number of divisors of the numbers 1 through 10, read column-by-column instead of row-by-row.

1.8 Remark — Reading the Grid by Columns Gives the Divisor Counts

The same 27 dots can be tallied down the columns instead of across the rows. Column b contains a dot for each divisor a of b , so its height is $d(b)$, the number-of-divisors function:

b	divisors a	$d(b)$
1	1	1
2	1, 2	2
4, 9	e.g. 1, 2, 4 / 1, 3, 9	3
6, 8, 10	e.g. 1, 2, 3, 6	4

Table 1.2

Prime columns ($b = 2, 3, 5, 7$) have height 2; the perfect squares 4, 9 have the odd height 3 (divisors pair up except the square root) — a small echo of the divisor-counting ideas that recur throughout number theory. Rows and columns give two different book-keepings of the *same* relation, and both must total 27.